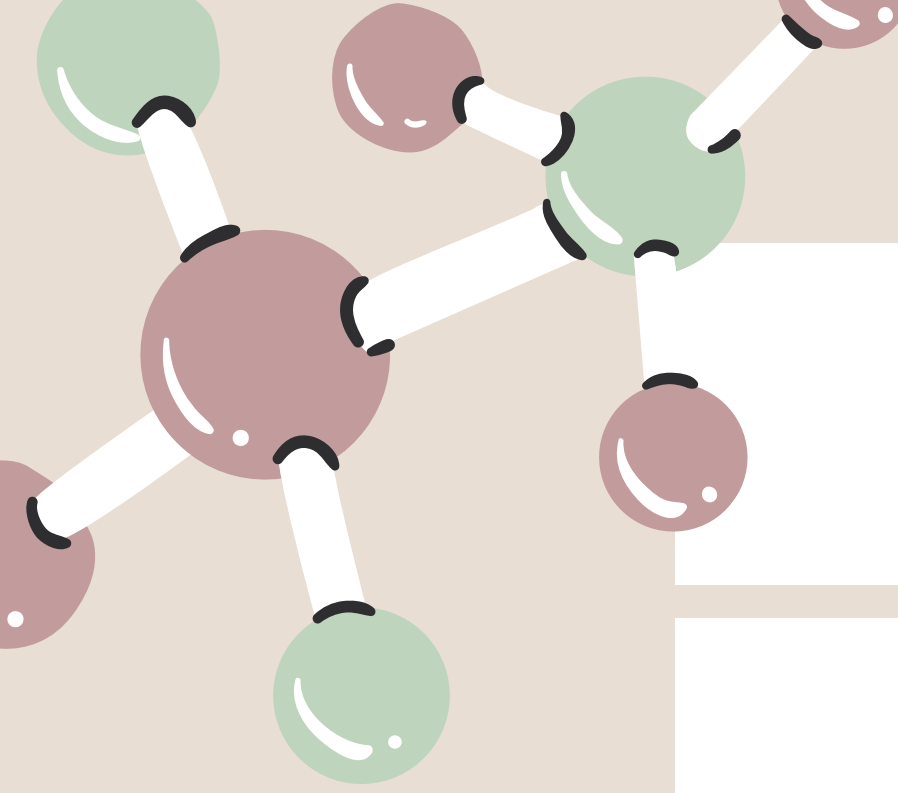




Invisible Ink



LESSON

This lab demonstrates chemical reactions and specifically introduces the concepts of carbonization, oxidation, and pH levels. This lab teaches students about how specific chemicals have visible reactions to being exposed to acidity, especially when a strong acid reacts with a strong base (alkaline) substance. Specifically, when the lab is done with grape juice and baking soda, the former acts as an indicator (something that changes color when detecting acid levels), while the latter is a base.



Both variations provide hands-on, historically accurate demonstrations of invisible ink using everyday materials, resulting in an engaging and educational experiment for students!

EXPERIMENT

Baking Soda & Grape Juice

Equal parts baking soda and water are mixed, then used to paint messages onto white paper. After drying, grape juice (usually concentrate or thawed) is brushed over the message, causing a pH reaction that permanently reveals the message.

Fun fact: soldiers during the Revolutionary War and World War I used this method to send secret messages hidden in regular letters!